

University of the Armed Forces - ESPE
TEST CASES:

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ID	Module	Function	Description	Input / Condition	Expected Result
TC01	Module	Constructor	Register resident WITH parking	Valid data + assignedParkingSpace "A10"	Resident created; userType = WITH_PARKING
TC02	Resident	Constructor	Register resident WITHOUT parking	Valid data; no space assigned	userType = ROTATING
TC03	Resident	Validation	Register resident with invalid email	Email missing '@'	Operation rejected
TC04	Resident	Validation	Register resident with duplicate ID	residentID already exists	Resident not created
TC05	Resident	addVehicle()	Add first vehicle to resident	Plate "ABC111"	Vehicle added; returns true
TC06	Resident	addVehicle()	Add duplicate plate	Plate already exists	returns false
TC07	Resident	addVehicle()	Add vehicle with null plate	plate = null	Vehicle rejected

TC08	Resident	removeVehicle()	Remove existing vehicle	Remove “ABC111”	returns true
TC09	Resident	removeVehicle()	Remove non-existing plate	“XXX000” not found	returns false
TC10	Resident	authorizeVisitor()	Add new authorized visitor	visitorID “V01”	Visitor added; returns true
TC11	Resident	authorizeVisitor()	Duplicate authorized visitor	“V01” already exists	returns false
TC12	Resident	removeAuthorizedVisitor()	Remove existing visitor	remove “V01”	returns true
TC13	Resident	removeAuthorizedVisitor()	Remove non-existing visitor	visitorID not in list	returns false
TC14	Resident	setCurrentRental()	Assign active rental to ROTATING resident	Valid Rental object	currentRental saved
TC15	Resident	hasActiveRental()	Check rental after assigning	rental.isActive() == true	returns true
TC16	Resident	setCurrentRental()	End rental (set null)	currentRental = null	hasActiveRental() returns false
TC17	Resident	getRentedSpace()	Get rented space ID	rental.spaceId = “P20”	returns “P20”

TC18	Resident	getResidentInfo()	Display info WITH assigned parking	userType = WITH_PARKING	Shows parking space
TC19	Resident	getResidentInfo()	Display info WITH active rental	ROTATING + rental active	Shows rental space + end date
TC20	Resident	getResidentInfo()	Resident with 0 vehicles	vehicles.size = 0	Shows 'Vehicles: 0'
TC21	Resident	getResidentInfo()	Resident with multiple vehicles	vehicle list size = 3	Displays 3 items in list
TC22	Resident	setEmail()	Update email field	newEmail valid	Email updated
TC23	Resident	setPhone()	Update phone field	valid phone	Phone updated
TC24	Resident	getResidents()	Read full resident list	JSON loaded; size=33	Returns list successfully
TC25	ResidentData	setResidents()	Replace resident list	new list of 10 residents	List updated; size = 10
TC26	ResidentManager	addResident	Add a valid new resident (integration).	Valid Resident object (no duplicate ID).	Returns true. getTotalResidents() increments.
TC27	ResidentManager	findResidentById	Search for a non-existent resident.	residentID = "R999".	Returns null.

TC28	ResidentManager	findResidentByVehiclePlate	Find resident by an existing vehicle plate.	Plate ABC111.	Returns the owner Resident object.
TC29	ResidentManager	findResidentByVehiclePlate	Search for resident by a non-existent vehicle plate.	Plate NON-EXIST.	Returns null.
TC30	ResidentManager	removeResident	Remove an **existing** resident.	residentID = "R123".	Returns true. getTotalResidents() decrements.
TC31	ResidentManager	removeResident	Attempt to remove a non-existent resident.	residentID = "R999".	Returns false. getTotalResidents() is unchanged.
TC32	ResidentManager	updateResidentInfo	Update info for a non-existent resident.	residentID = "R999", new data.	Returns false. No data is modified.
TC33	ResidentManager	getResidentsWithParking	Filter residents with assigned parking.	Mixed resident list.	Returns list containing only userType = WITH_PARKING.
TC34	ResidentManager	getRotatingResidents	Filter residents of type Rotating.	Mixed resident list.	Returns list containing only userType = ROTATING.
TC35	ResidentManager	createRentalForResident	Create a successful rental for a ROTATING resident.	ROTATING resident ID, available space B20, 3 months.	Returns Rental object. Resident has hasActiveRental() == true.
TC36	ResidentManager	createRentalForResident	Attempt to create rental for WITH_PARKING resident.	WITH_PARKING resident ID.	Returns null. Rejected due to user type.
TC37	ResidentManager	createRentalForResident	Attempt to create rental when one is already active.	ROTATING resident with	Returns null. Rejected due to active rental.

				hasActiveRental() == true.	
TC38	ResidentManager	renewRentalForResident	Successfully renew an active rental.	ID with active rental, additionalMonths=6.	Returns true. Rental's endDate is extended.
TC39	ResidentManager	cancelRentalForResident	Cancel active rental and free space.	ID with active rental.	Returns true. hasActiveRental() is false. Space updated to AVAILABLE in ParkingLot.
TC40	ParkingLot	findAvailableSpace	Search for and find an available space.	Lot has at least one AVAILABLE space.	Returns ParkingSpaceLibrary object with AVAILABLE status.
TC41	ParkingLot	findAvailableSpace	Attempt to find space when the lot is full.	All spaces are marked as	Returns null.
TC42	ParkingLot	assignSpaceToVehicle	Manually assign a plate to an existing and available space.	spaceId="A15", plate="XYZ999", type="Manual"	Returns true. Space A15 is OCCUPIED by XYZ999.
TC43	ParkingLot	findSpaceByVehicle	Search for space by occupying plate.	Plate XYZ999 (from TC42).	Returns the ParkingSpaceLibrary object for space A15.
TC44	SecurityGuard	registerEntry	Register entry for an authorized Visitor (main flow).	Visitor plate V01, currentTime.	System calls processVisitorEntry, and entry is successfully logged.
TC45	SecurityGuard	registerExit	Register exit for a Visitor with an active pass.	Visitor plate V01, currentTime.	System calls recordExit. Temporary

					pass is set to inactive.
TC46	VisitorManager	findVisitorById	Search for an existing visitor.	visitorID = "V01".	Returns Visitor object.
TC47	VisitorManager	findVisitorById	Search for a non-existent visitor.	visitorID = "V999".	Returns null.
TC48	VisitorManager	addVisitor	Add a new visitor (non-existent).	Visitor object with unique ID.	Visitor is added. saveVisitors() is called.
TC49	VisitorManager	addVisitor	Attempt to add visitor with a duplicate ID.	Visitor object with an existing ID.	Operation ignored/rejected. saveVisitors() is not called.
TC50	ResidentManager	generateResidentsReport	Generate the full Residents Report.	Execute the method.	Returns String containing: Title, totals (with/without parking), and detailed info for each resident.
TC51	LicensePlate	validatePlate()	Valid plate format (3 letters + 3 digits).	ABC123	Returns true
TC52	LicensePlate	validatePlate()	Invalid short plate.	AB12	Returns false
TC53	LicensePlate	validatePlate()	Plate with special characters.	AB#123	Returns false

TC54	EntryExitRecord	registerEntry()	Record entry for known resident with vehicle registered.	plate = "XYZ999", type=Resident.	Entry record saved with timestamp; active=true.
TC55	EntryExitRecord	registerExit()	Attempt exit without previous entry.	plate not found in active records.	Exit rejected; returns false.
TC56	EntryExitRecord	closeRecord()	Close entry record normally.	Active record exists for "XYZ999".	endTime filled; active=false
TC57	ParkingStructure	loadBlocks()	Load building blocks from JSON	JSON contains 3 blocks.	Structure size = 3; all blocks loaded successfully.
TC58	ParkingZone	getZoneById()	Retrieve the existing parking zone.	zoneId = "Z1".	Returns Zone object; spaces list accessible.
TC59	ParkingZone	getZoneById()	Search non-existent zone	zoneId = "Z99".	Returns null.
TC60	ParkingSpace / SpaceDefinition	isAvailable()	Space initially free	status = AVAILABLE.	Returns true.
TC61	ParkingSpace / SpaceDefinition	occupy()	Occupy available space.	plate="AAA111"; current status = AVAILABLE.	status = OCCUPIED; plate saved.
TC62	ParkingSpace /	freeSpace()	Free occupied space.	Space currently OCCUPIED	status = AVAILABLE; plate cleared.

	SpaceDefini tion				
TC63	ParkingLot	countAvailableSpaces()	Count free spaces.	50 free of 173.	Returns 50
TC64	ParkingLot	countOccupiedSpaces()	Count occupied spaces.	123 occupied.	Returns 123
TC65	Vehicle	setPlate()	Change the plate to a valid one.	BBB222	Plate updated correctly.
TC66	Vehicle	setPlate()	Change plate to invalid format.	11BB22”.	Operation rejected; plate not changed.
TC67	Visitor	setExpirationDate()	Update visitor pass expiration.	new date = +3 days.	Field updated.
TC68	Visitor	isPassActive()	Check valid pass.	expirationDate today+1.	Returns true.
TC69	Visitor	isPassActive()	Check expired pass.	expirationDate = yesterday.	Returns false.
TC70	JsonDataManager	loadResidentsFromJson()	File exists and valid.	residents.json correct.	Returns list; size matches file
TC71	JsonDataManager	loadResidentsFromJson()	Missing file.	File not found.	Returns an empty list or throws a handled exception.
TC72	JsonDataManager	saveParkingLot()	Save parking lot state.	Valid ParkingLot object.	JSON updated; returns true.

TC73	ParkingControlSystem	initializeSystem()	Full system load success	All JSON files valid.	system.active = true; managers initialized.
TC74	ParkingControlSystem	initializeSystem()	Fail to load residents JSON.	residents.json corrupted.	active=false; error logged.
TC75	ParkingControlSystem	shutdown()	End system execution.	User selects "Salir".	active=false; resources released; goodbye message.
TC76	ParkingLot	getParkingLotStatus()	Generate textual summary of lot occupancy	Mixed occupied and available spaces	Returns non-empty String with counts and list formatting
TC77	ParkingLot	freeSpaceByPlate()	Free a space occupied by a known plate	plate "DEF-333" currently assigned	Space changes to AVAILABLE; method returns true
TC78	ParkingLot	freeSpaceByPlate()	Free a space for an unknown plate	plate "NOPE-000" not in any occupied slot	Operation rejected; returns false
TC79	ParkingLot	getSpacesByStatus()	Retrieve list of spaces with AVAILABLE status	Parking lot partially filled	Returns list.size ≥ 1; items status = AVAILABLE
TC80	ResidentManager	getTotalResidents()	Count total residents after multiple insertions	Add 5 new residents	getTotalResidents() returns 5
TC81	ResidentManager	updateResidentInfo()	Update phone and email of existing resident	New valid phone + email	Resident fields updated successfully; returns true

TC82	ResidentMa nager	updateResidentInfo()	Provide invalid email when updating	email missing “@”	Update rejected; returns false
TC83	ResidentMa nager	getAllResidentsSum mary()	Generate short summary lines for each resident	Resident list contains 3 entries	Summary returns 3 formatted lines
TC84	Resident	getVehicleCount()	Count vehicles linked to resident	Resident has 2 vehicles	getVehicleCou nt() = 2
TC85	Resident	removeAllVehicles()	Remove all vehicles from resident	Resident with 3 vehicles	Vehicle list cleared; size = 0
TC86	Resident	setName()	Update resident name	name = “Luis Rivera”	Name updated correctly
TC87	Rental	calculateRemaining Days()	Determine days until expiration	End date = today + 20	Returns 20
TC88	Rental	cancel()	Cancel an active rental	Rental active	active=false; endDate set to today
TC89	Vehicle	equals()	Check equality based on plate	Two vehicles with same plate	equals() = true
TC90	Vehicle	clone()	Duplicate a vehicle object	Vehicle with plate “PQR-111”	New instance with same plate & data
TC91	Visitor	setName()	Update visitor name	New name “Daniel Soto”	Visitor name updated successfully
TC92	Visitor	validatePass()	Validate pass and return reason if invalid	expirationDate in past	Returns “Expired pass”
TC93	SecurityGu ard	verifyResidentPlate()	Check if a plate belongs to any resident	plate registered in system	match = true

